

Overview

i2ai Team Project Instructions

Learning Objectives

- Apply coordination, collaboration and communication skills in a team-based client engagement.
- Gather and analyze client requirements.
- Critically apply agentic AI skills and tools to solve a real-world problem based on client requirements.

Overview

Each team in the Intro to AI course (Fall 2026) at CMU's Heinz College will complete a semester-long project for a client. This is a low-stakes (no contract, no payment, no legal obligations) project, NOT a capstone project.

A client provides a small and actionable problem for students to address using agentic AI. The client may or may not provide access to the organization's data. Students may build an app, a website, or other solutions using simulated data/accounts. The project list and team assignments are available [here](#).

A client may be assigned one or two student teams. We do so for two reasons:

1. Keep the team size at 5 or less. Teams bigger than 5 become very difficult to manage/coordinate.
2. Different teams can approach the same problem from different angles, and the client can get two different solutions that they can compare and contrast. The two teams can learn new ideas from each other as well (during the final presentations).

Technical Requirements

In addition to meeting client requirements, the project team must demonstrate:

1. How the team has accomplished the project using an agentic workflow, and/or have helped the client establish an agentic workflow
2. How a pretrained model (e.g. an LLM, a pretrained prediction/vision model, etc), or a model that the team has trained from scratch, is incorporated into the solution
3. At least five agentic tools (with well-written docstring documentation designed for LLM agents to read) that the team has custom designed for this project and for an LLM-based agent to use and to control the LLM-based agent's behavior in a more deterministic manner
4. A skill folder with a skill.md file and/or templates that the team used for developing the solution, or that the team has designed for the client to use in the future
5. Methods that the team has used to evaluate the system's performance
6. Methods that the team has used to protect client data, or to simulate test data. If you simulated data, explain how you ensure the quality of the simulated dataset.

7. Safety guardrails (perhaps as agentic tools from 3., or parts of the skill folder, etc) to address potential risks and ethical concerns
 - a. How the model behaves with vs. without the safety guardrails to illustrate the effectiveness of such safety guardrails

Project Privacy & Confidentiality

The nature of the projects and client identity is by default confidential. Some clients may ask you to sign an NDA. Do NOT discuss project or client identity details without permission. Always check with the client and get permission before you share details with those outside the project/TA/instructor. To post about the project on social media (e.g. LinkedIn), send a draft to your client for approval before posting.

Data Access

Most clients won't be able to share sensitive data with you. You will be responsible to simulate data for testing as needed.

Intellectual Property

The intellectual property of this work is shared between the student team and the client.

Client Interaction

Three client meetings are required during the semester:

- (1) first client requirements meeting: late September
- (2) progress update: early November
- (3) final presentation: early December

Deliverables - See syllabus for due dates

Team Project #1: Team Contract and Project Topic Exploration

Team Project #2: Progress update

Team Project #3: Final Deliverable

Team Project #1

Team Project #1: Team Contract & Scope

Purpose

Before you engage a client, your team needs (1) a working agreement on how you'll collaborate, and (2) a claimed topic/client with an initial understanding of the problem space.

Part A - Team Contract

Complete the [team contract template](#) covering:

- Team member contact info, availability, strengths, and areas to watch
- Team norms (meeting cadence, communication, decision-making)
- Expectations and conflict resolution process
- Team goals

Part B - Client Meeting Prep

1. Review information about the client assigned to your team.
2. Send an initial outreach email to your client (instructions provided in the Appendix) to introduce your team and request a kickoff meeting.
3. Research client organization and client information before the client meeting.
4. Hold your first client requirements meeting (late September, before Team Project #1 deliverable is due - see syllabus for the due date). Come prepared to ask about:
 - A specific, actionable problem they'd like addressed
 - Constraints, context, or examples that help scope the problem
 - What data (simulated/public) you'll need, and whether an NDA is required
 - What a successful outcome looks like to them by December
 - A date/time for the progress update meeting: late October or early November
5. Invite the client to attend your final poster presentation (see syllabus for date) - discuss in the meeting and follow up with a calendar invite.
 - If your client is remote, schedule a time to present your final deliverables to the client.

Part C - Project Scope

Based on the client meeting, write a clear project scope document based on these [PMI instructions](#). Send the project scope document to your client for approval.

Build a Github kanban board to start managing your project. See here for an [example board](#). Follow online tutorials like [this one](#).

Submission in Canvas

Three documents:

1. Signed team contract
2. Short (~1 page) client meeting prep document: who the client is (or anonymized description if confidentiality requested), the problem as currently understood, and a list of questions for the client before heading into the first client meeting.
3. Project scope document (which your client may or may not have approved) with a link to your team's kanban board.

Appendix. Template email to the client

Instructions for teams:

1. Find out if more than one team is assigned to the same client. All teams assigned to the same client should work together with the assigned project TA to identify meeting times/dates that work for most people. Include these options in the email to the client for the client to select.
 - a. Note: Even if multiple teams are meeting with the same client, the teams will work on the same topic separately. Some clients may give different teams different topics, but most will give the same topic to all teams.
2. Fill in the bracketed placeholders, and adjust the tone to fit your client.
3. Appoint one person to send the email to the client (copying the TA, instructor, and all team members) to schedule your initial requirements meeting.

Subject: Intro & Scheduling – [Team Name] / CMU Intro to AI Project

Hi [Client Name],

My name is [Your Name], and I'm reaching out on behalf of our team(s) – [list team members] – from Carnegie Mellon University's Heinz College. We're students in the Intro to AI course this semester with Dr. Rachel Chung as our professor. We've been paired with you as our project client for a semester-long agentic AI project. Our project will be supported by our TA [TA name].

This is an academic project, with no contract, payment, or legal obligation on your end. We won't need access to your organization's actual data or systems, although you are welcome to

provide data or access. We will build our solution using simulated or publicly available data. If you'd like us to sign an NDA before discussing any details, we're glad to do that.

We'd love to set up a short kickoff call (30 minutes or so) before the end of September to understand:

- A specific, actionable problem you'd like us to explore using agentic AI (an AI system that can plan and take multi-step actions, not just answer questions)
- Any context, constraints, or examples that would help us scope the problem well
- What a useful outcome would look like from your perspective by early December

Would any of the following work for a quick call? [Option 1] / [Option 2] / [Option 3] (Specify time zone as the client may be in a different time zone.)

(For local clients only) Please let us know if you'd like to meet in person at CMU or your office, or remotely on Zoom.

Looking forward to working with you this semester.

Best, [Your Name] [Contact info]

contract

Team Contract — Intro to AI, Fall 2026

Purpose: Before you start your client project, your team needs a shared agreement on how you'll work together — who's good at what, when you're available, and how you'll handle it if something goes sideways. This isn't busywork: teams that skip this conversation tend to hit the same conflicts around week 6 that a 20-minute discussion now would have prevented.

Each team should:

1. Sit together and fill this out collaboratively — don't split it up and stitch it together later. The point is the conversation, not the document.
2. Be honest about weaknesses and availability. A contract full of "no weaknesses, available anytime" helps no one in week 10.
3. Agree on the norms and conflict resolution sections as a group — these are the parts you'll actually reference if something goes wrong.
4. Submit the completed contract as **Team Project #1**, alongside your topic selection.

1. Team Info

Team name

Client / project topic

2. Team Members

Repeat this block for each member.

Name: **Email:** **Phone (optional, for urgent coordination):** **Availability/commitment:**

(recurring conflicts — classes, work, sports — and general flexibility) **Strengths:** (skills, tools, or working styles you bring) **Areas to watch:** (where you might need support or accountability from teammates — e.g., procrastination, limited coding background, tends to overcommit) **Preferred role(s) on this project:** (e.g., client communication, technical build, presentation/design, project management)

3. Team Norms

Meeting cadence: (how often, what day/time, in-person or virtual) **Check-in method:** (Slack, group chat, etc. — and expected response time) **Decision-making:** (how will the team resolve disagreements on approach — majority vote, client-facing member decides, escalate to

TA/instructor?) **Work distribution:** (how will tasks get assigned and tracked — e.g., GitHub project board)

4. Expectations

What does "doing your share" look like for this team?

Attendance: (what happens if someone misses a meeting or a deadline)

5. Conflict Resolution

If a team member isn't meeting expectations, what's the process before escalating to the TA/instructor?

At what point do you escalate, and to whom?

6. Team Goals

What does a successful outcome look like for this team, beyond just the grade?

Signatures / acknowledgment *By signing below, each member confirms they've discussed and agreed to the above.*

Name	Signature	Date
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Name	Signature	Date
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Name	Signature	Date
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Name	Signature	Date
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Name	Signature	Date
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rubric TP1

TP1 Rubric - Team Contract & Scope

Total: 30 points

TBD

draft

Team Project #2

Team Project #2: Progress Update

Purpose

Check in on both the client relationship and the technical build roughly halfway through the project, catch scope or coordination problems early, and give teams structured feedback before the final push.

What to Prepare

1. **Progress against plan:** what you scoped at TP1 vs. where you actually are.
2. **Working prototype or partial demo:** even an early, rough agentic AI component — this isn't the polished final product.
3. **Blockers/open questions:** technical, client-access, or team-coordination issues you need help with.
4. **Updated scope (if changed):** what shifted since TP1 and why.
5. **Project board snapshot:** current state of your GitHub kanban board (done/in progress/to-do).

Client Meeting

Present the above to your client in an early-November progress-update meeting. Come away with explicit confirmation (or correction) of direction for the final stretch.

Submission

- Short progress report (1-3 pages) covering the five points above
- Link to current GitHub repo state (code + project board)

rubric TP2

TP2 Rubric - Progress Update

Total: 50 points

Criteria	Points	What we're looking for
Progress vs. plan	10	Honest, specific comparison of TP1 scope to actual progress; meaningful work completed by this point
Working prototype/demo	12	A functioning (even if rough) agentic AI component – real progress on the actual build, not just planning docs
Blockers & problem-solving	8	Blockers are clearly identified with a credible plan to resolve them
Scope updates	5	If scope changed, the reasoning is sound and clearly explained
Project board upkeep	5	Kanban board reflects real, current task status
Client meeting quality	10	Client meeting held; team came away with clear, confirmed direction for the remainder of the project

Team Project #3

Team Project #3: Final Deliverable

Purpose

Present a working prototype of an agentic AI application solving your client's problem, hosted in a complete GitHub repo, to faculty, guests, and students.

Required GitHub Repo Contents

1. **README** — comprehensive project report addressing all technical requirements given in the project overview, including:
 - Visuals illustrating your core concept and its business relevance
 - Author list with links to each member's GitHub profile (with a proper bio)
 - Project scope — as narrowly and specifically defined as possible
 - Project details — clearly organized, concise writeups
 - "What's next?" — future development directions and open concerns
 - Responsible AI considerations
 - Reference list, including **at least one** high-quality research paper (not news/blog/LinkedIn), published within the last 3 years
 - Do not disclose your client's identity without their permission
2. **Runnable demo code** — a .py file, Jupyter notebook, or Colab link. Must run live during the presentation. Cite any code sourced from elsewhere.
3. **Externally hosted app/website** — built using GenAI tools with agentic capabilities (e.g., an agent that plans/acts across steps, not just single-turn Q&A).
4. **Project board** — kanban-style board (done / in progress / to-do) used throughout the semester. Be ready to discuss how your team actually used it.

Presentation Requirements

Be ready to discuss:

1. **Problem statement** — what you solved and why it matters
2. **Research paper** — the required paper and how it informed your approach
3. **Code demo** — keep it simple; illustrate one essential concept rather than the whole project
4. **Website/app demo** — show it live and explain its agentic design
 - a. Address all of the technical requirements given in the overview

rubric TP3

TP3 Rubric - Final Deliverable

Total: 120

For each rubric: Excellent = 100% Satisfactory = 75% Unsatisfactory = 50% Poor = 25%

Criteria	Percentage
Code demo: Technically challenging, logically sound, and thoughtfully put together.	7%
Website demo: Creative and thoughtful use of vibe coding	7%
Poster is visually effective and covers everything that's required for the project.	7%
Github repo is organized and covers everything required for the project effectively.	7%
Github project kanban board is visually organized.	7%
Research paper is well chosen and well discussed verbally and on the poster.	7%
References are cited properly.	7%
The team articulated their problem statement, rationale, and overall approach effectively to the judges.	9%
The team articulated the business opportunities and challenges of their project work clearly and effectively to the judges.	9%
The team explained the technical components of their project clearly and effectively for a non-technical audience.	12%
The team helped the judges learned a significant amount of new knowledge about AI.	7%

The team demonstrated creativity in their approach to their project topic.	7%
The team demonstrates effective team collaboration with all team members actively engaged in presentation and Q&A.	7%